Genetic Algorithm Research Project

Daniel Jones, Youngrak Kim, Haocheng Tang, & Helen Xu

Rose-Hulman Institute of Technology, CSSE220

[2/21/2025]

1 Introduction

[Explain what this project is about.]

- This project is about creating chromosomes and mutating their genotypes based on different parameters and seeing how the different parameters affect the change of the genotypes of a population over time.

[What is the motivation for the work you did?]

- The GARP project seemed more interesting than the arcade game.

[Why is it important or relevant?]

- Genetic Algorithms are an effective way to solve new complex problems by exploring multiple solutions simultaneously and finding a more global solution rather than a local maximum.

2 Background

[What is a genetic algorithm?]

- Genetic algorithms are a class of optimization and search techniques. They are a way to simulate biological evolution. It is helpful and effective for solving complex problems instead of using traditional methods.

[Provide anything useful the reader should know about your project.]

- It is a biological simulation program. You can input the chromosome size and the length of chromosome to present everyone. The chromosome's bit string is randomly generated. Like the environment, the chromosomes might mutate as the time passes. Also, some chromosomes might crossover, which presents the reproduction of two animals. Moreover, elitism is another feature of animals. Animals tend to reproduce with the ones with good genes. This program also provides some functions to judge whether the chromosome is good or bad and some different ways to select the chromosomes.

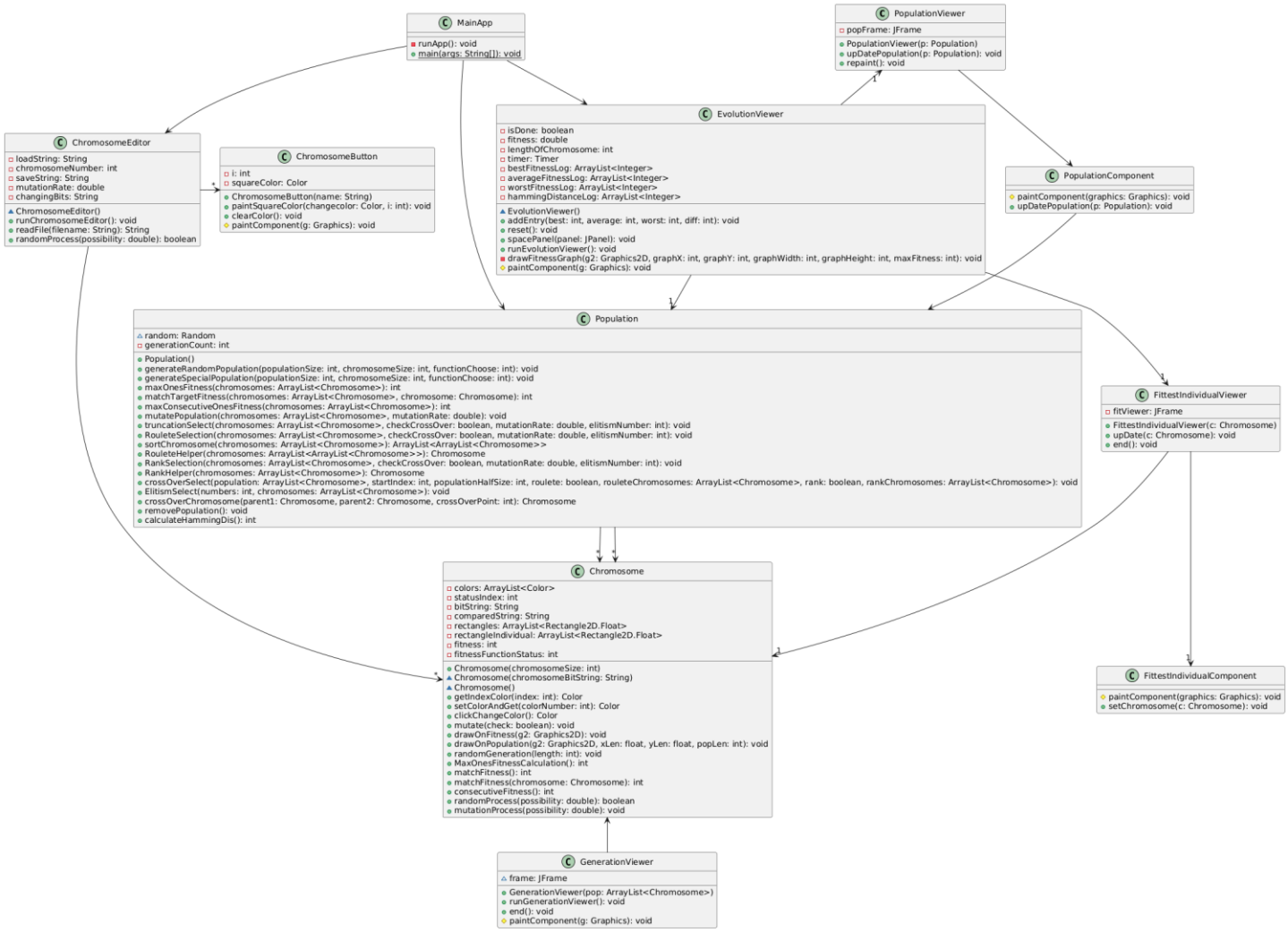
3 Methods

[How did you accomplish this work?]

- We separate the work from each other, searching for information we need to fulfill the requirements. If anyone was having trouble, they reached out and someone would step in and either debug or they would collaboratively figure out a solution to the problem.

[Provide a UML diagram of part or the whole system.]

plantuml diagram



[Explain how the experiments were run.]

- When we run the program, MainApp class will execute the runApp method, which has runEvolutionViewer method, then evolution viewer will be called, and new window will appear. After filling in all parameter text fields and checking box for crossover, special population in the user interface, the population class will be called from given parameters, and other Chromosome class, PopulationViewer class, and FittestIndividualViewer will be drawn.

4 Selection Option Science

[Which option did you select and why?]

- We chose science topic, and this is because we are interested in application of java into genetic algorithms, and especially science part, we want to approach biological perspective through genetic algorithms.

[How does the bitstring relate to your specific phenotype?]

- A bitstring can represent a specific phenotype by encoding genetic information in a digital format. Each bit (0 or 1) in the bitstring can correspond to the presence or absence of a particular genetic marker, allele, or characteristic associated with the phenotype. The mapping between the bitstring and the phenotype depends on the encoding scheme used.

[How well did you accomplish this goal?]

- We did well on mutating, crossover and elitism simulating the biology evolution, we successfully visualized our population dynamics, fitness trend functions, population, and best fittest chromosome. Also, application of binary fitness function helps our experiment about genetic algorithm with efficiency and verify cause and effect of Baldwin effect.

5 Results

We based our project and experiments on the article "How Learning Can Guide Evolution". Its main idea is that adaptive learning guides evolution, also known as the Baldwin effect, which allows learning organisms to evolve more efficiently than non-learning ones by creating evolutionary pathways that would otherwise be difficult to discover. The population consisted of chromosomes that will have increased fitness for every second due to selection method and other features such as mutation, crossover, and elitism, and it will eventually reach the max fitness after series of reproducing generation.

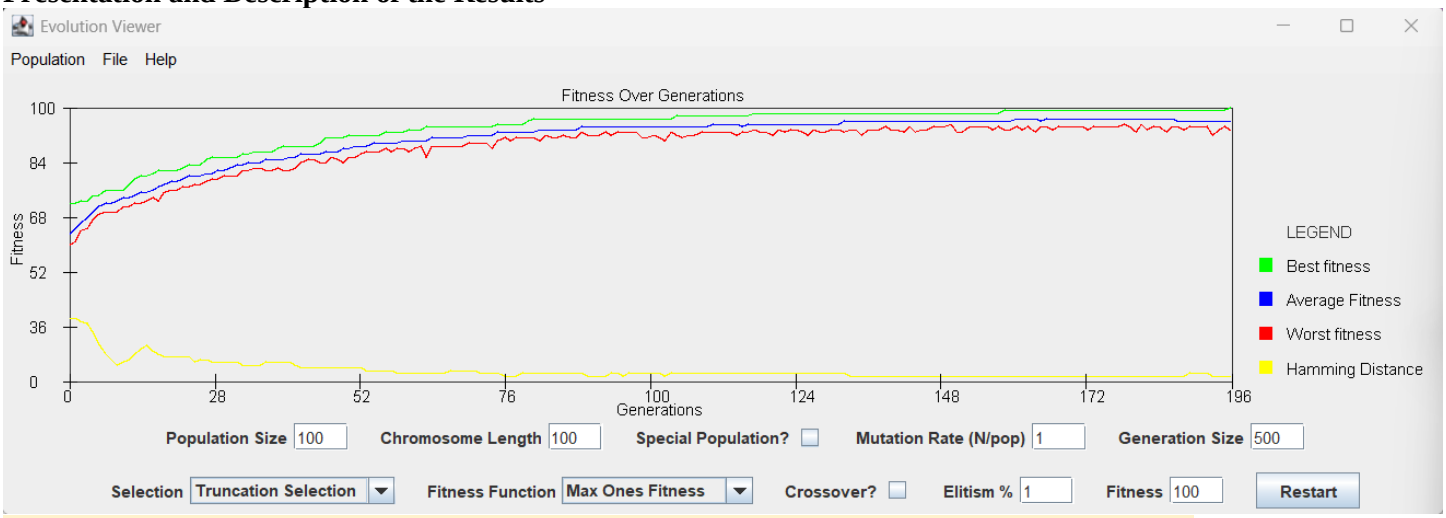
Experiment 1 – Evolution with Mutation Only

Experiment Description and Parameters
This experiment is designed to observe and measure the rate and variability in evolution over multiple runs.

population size:	100		*elitism*	1
chromosome length:	100		*selection method:*	truncation
max generations:	500		*crossover method:*	disable
fitness function	MaxOnes		*mutation rate:*	1%

Hypothesis
[We expect to see evolving generations from a random population with an increasing trend of fitness through truncation selection and mutation based on adaptive learning. Also, by selecting each population from truncation selection method, hamming distance must be approaching almost 0 by the last generation. During evolution, the fitness tendency must keep increasing and done before reaching maximum generation 500.]

Presentation and Description of the Results



[This experiment showed a perfect solution in 196 generations. You can see an increasing trend line.]

run #	1	2	3	4	5	6	7	8	9	10
gens	*205*	*237*	*222*	*210*	*218*	*204*	*196*	*214*	*191*	*235*

[The variation from run to run was high with a mean of 213.2 and a standard deviation of 14.4. This aligns with my hypothesis.]

What if anything you can conclude and anything you learned or affirmed by doing so?

[Based on the data collected, it is safe to conclude that adaptive processes within the organism can be very effective in guiding evolution which supports the Baldwin effect by showing increasing fitness from truncation leads to learning the parent's superior gene and implementing it to the next generation's gene.]
[In the future it would be interesting to explore about how adaptive process is influenced from guiding evolution, which occurs in different conditions.]

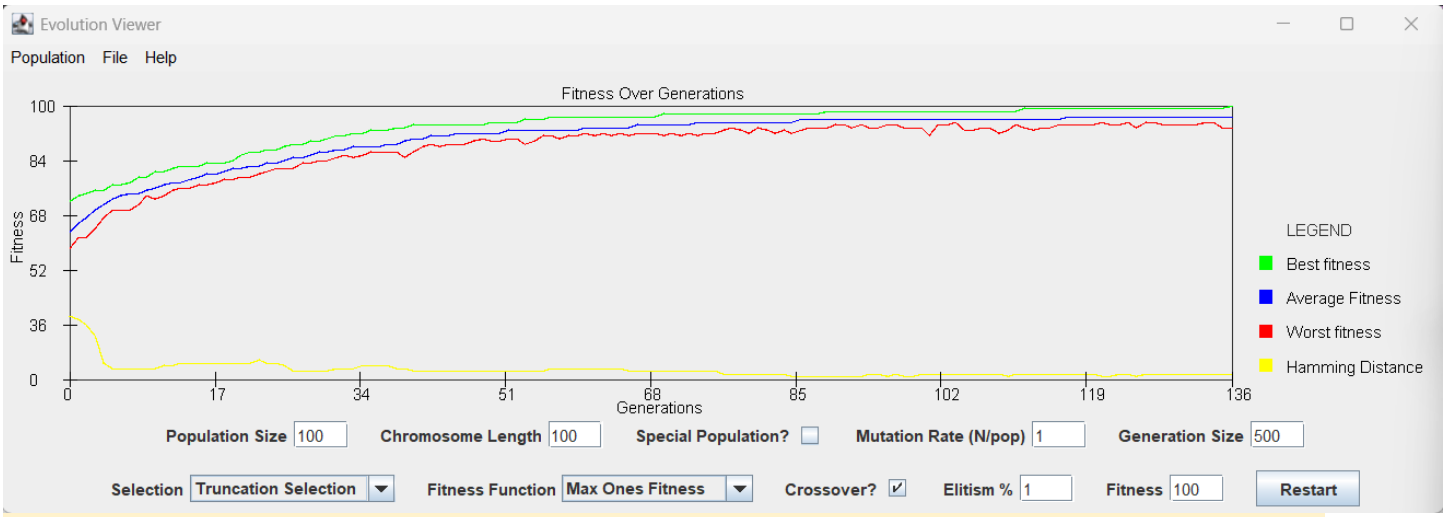
Experiment 2 – Evolution with Mutation and Crossover

Experiment Description and Parameters
This experiment is designed to observe and measure the rate and variability in evolution over multiple runs when crossover is included in addition to mutation.

population size:	100		*elitism*	1
chromosome length:	100		*selection method:*	truncation
max generations:	500		*crossover method:*	single-point
fitness function	MaxOnes		*mutation rate:*	1%

Hypothesis
[We expect to see evolving generations from a random population through truncation selection and mutation that has similar tendencies to experiment 1, without crossover. Additionally, for this experiment, the evolution by adaptive learning must be done in earlier generations with lower variation due to the crossover from their parent chromosomes, which leads to faster and stable reach to the fittest population.]

Presentation and Description of the Results



[This experiment showed a perfect solution in 136 generations. You can see almost same tendency comparing to experiment 1 in fitness and hamming distance.]

run #	1	2	3	4	5	6	7	8	9	10
gens	*135*	*122*	*121*	*133*	*136*	*119*	*128*	*135*	*140*	*139*

[The variation from run to run was low as expected with a mean of 130.8 and a standard deviation of 7.3. This aligns with my hypothesis that crossover affects guiding evolution.]

What if anything you can conclude and anything you learned or affirmed by doing so?

[Based on the data collected it is safe to conclude that it clearly demonstrates the theory of genetic algorithm that learning can significantly accelerate evolution by reshaping the search space in which genetic evolution operates within the organism can be very effective in guiding evolution, and crossover, which leads enhancing guiding evolution by duplicating their parent's gene, describes impact of it well.]
[In the future it would be interesting to explore how other conditions such as mutation rate, selection method, or elitism affect to guiding evolution.]

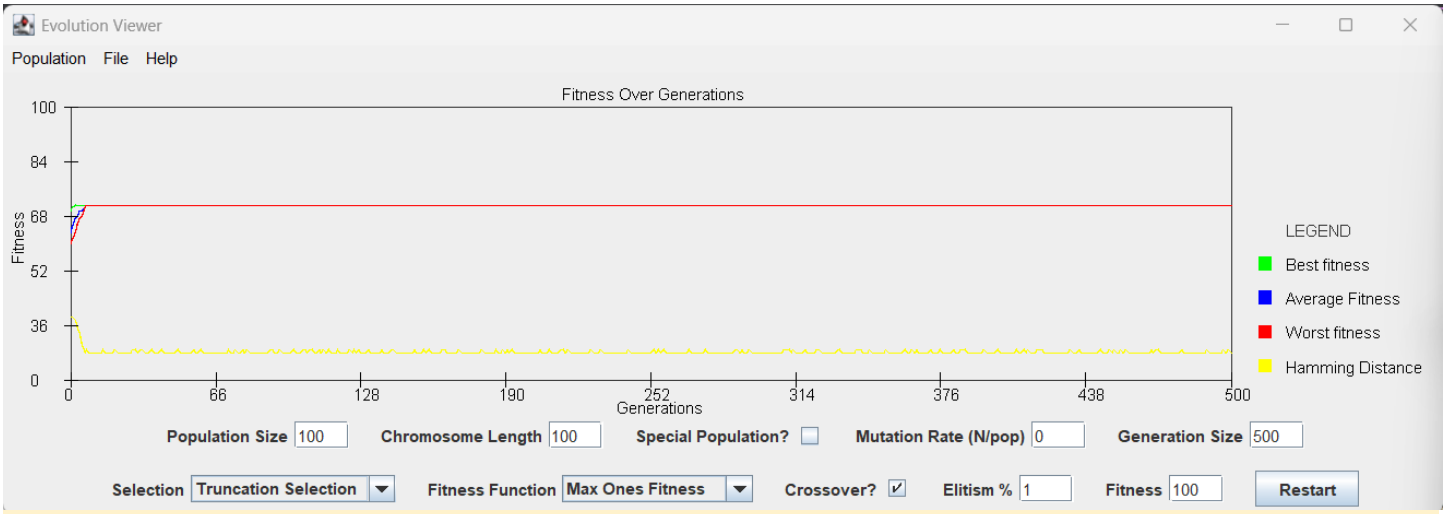
Experiment 3 – Evolution with Crossover Only

Experiment Description and Parameters
This experiment is designed to observe and measure the rate and variability in evolution over multiple runs when only crossover can be used to introduce variation into a population.

population size:	100		*elitism*	1
chromosome length:	100		*selection method:*	truncation
max generations:	500		*crossover method:*	single-point
fitness function	MaxOnes		*mutation rate:*	0%

Hypothesis
[We expect to see increasing trend of the fitness in population at first, but it will approach at some point and cannot go over until reaches maximum generations due to the 0% of mutation rate, which supports Baldwin Effect, which states that characteristics are not inherited which implies that what an organism learns during its lifetime cannot guide evolution. So, the final genes must be reached 500 for every test cases.]

Presentation and Description of the Results



[This experiment showed a perfect solution in 500 generations. You can see an increasing shift at first, but all three fitness are approaching approximately fitness of 6 until 500 generations are finished.]

run #	1	2	3	4	5	6	7	8	9	10
gens	*500*	*500*	*500*	*500*	*500*	*500*	*500*	*500*	*500*	*500*

[The variation from run to run was same as expected with a mean of 500 and a standard deviation of 0. This aligns with my hypothesis.]

What if anything you can conclude and anything you learned or affirmed by doing so?

[While not conclusive, results suggest that mutation rate affects significantly to adaptive learning through guiding evolution, by hampering population from producing new mutated chromosome and reproducing similar chromosome from their parents, eventually cannot exceed their parent's fitness, eventually confirms Baldwin effect.]
[In the future it would be interesting to explore what the effect would be with 100% mutation rate.]

Experiment 4 – Systematic Exploration of Evolution with Rank Selection

Experiment Description and Parameters
This experiment is designed to observe and measure the rate and variability in evolution over multiple runs when rank selection is selected instead of truncation selection, which can be used to introduce variation into a population.

population size:	100		*elitism*	1
chromosome length:	100		*selection method:*	Rank Selection
max generations:	500		*crossover method:*	disable
fitness function	MaxOnes		*mutation rate:*	1%

Hypothesis
[We expect to see this experiment indicates similar tendency to the truncation selection with same conditions as performed in experiment 1. In this experiment, because of the logic of ranked selection, which leads to superior selection than truncation, population will reach maximum fitness in earlier generation rather than truncation with less variation. This will support article that incorporating learning into evolution reduces the number of generations required to reach optimal adaptations.]

Presentation and Description of the Results



run #	1	2	3	4	5	6	7	8	9	10
gens	*147*	*158*	*151*	*150*	*162*	*150*	*152*	*149*	*151*	*163*

[The variation from run to run was same as expected with a mean of 153.3 and a standard deviation of 5.3. This aligns with my hypothesis.]

Discussion

[Based on the data collected it is safe to conclude that rank selection is more suitable than truncation selection for guiding evolution perspective, emphasizing the crucial role of adaptive processes in guiding genetic evolution. Because rank selection selects appropriate numbers of chromosomes based on their fitness compared to truncation selection which selects half of the population. Therefore, it is considered that rank selection is more applicable for genetic algorithm projects.]

[In the future it would be interesting to explore which selection method is best to get maximum fitness with less generation, and determining which algorithm of selection method affects to select population in terms of guiding evolution.]

Experiment 5 – Special Phenotype Evolution

Experiment Description and Parameters

This experiment is designed to see if different size populations with all other settings get a fitness of 10 at a similar number of generations

population size:	10/50/100/500/1000		*elitism*	20%
chromosome length:	100		*selection method:*	Binary Fitness
max generations:	1000		*crossover method:*	No
fitness function	10		*mutation rate:*	1

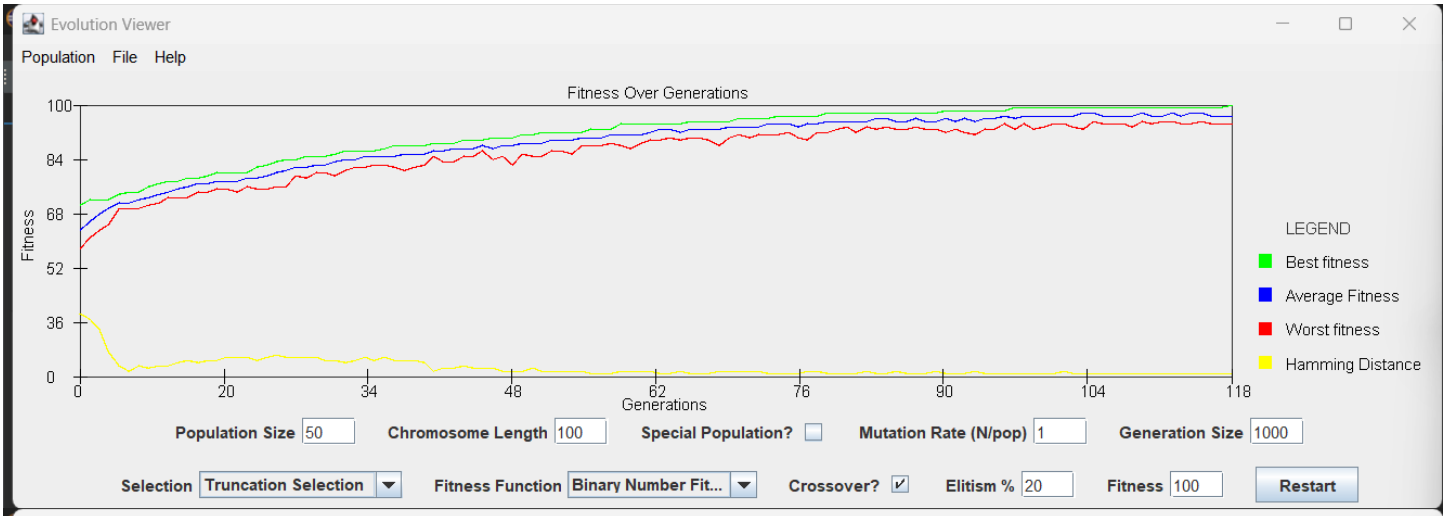
Hypothesis

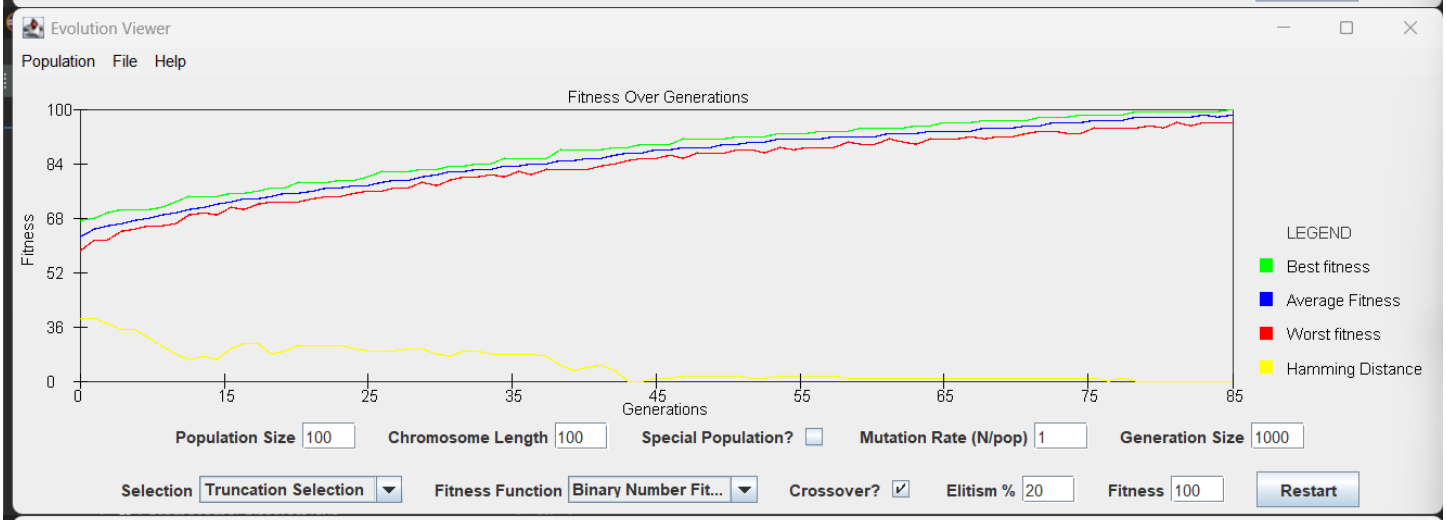
We hypothesized that as the population size went up the number of generations required to achieve set fitness would remain roughly the same.

Presentation and Description of the Results:

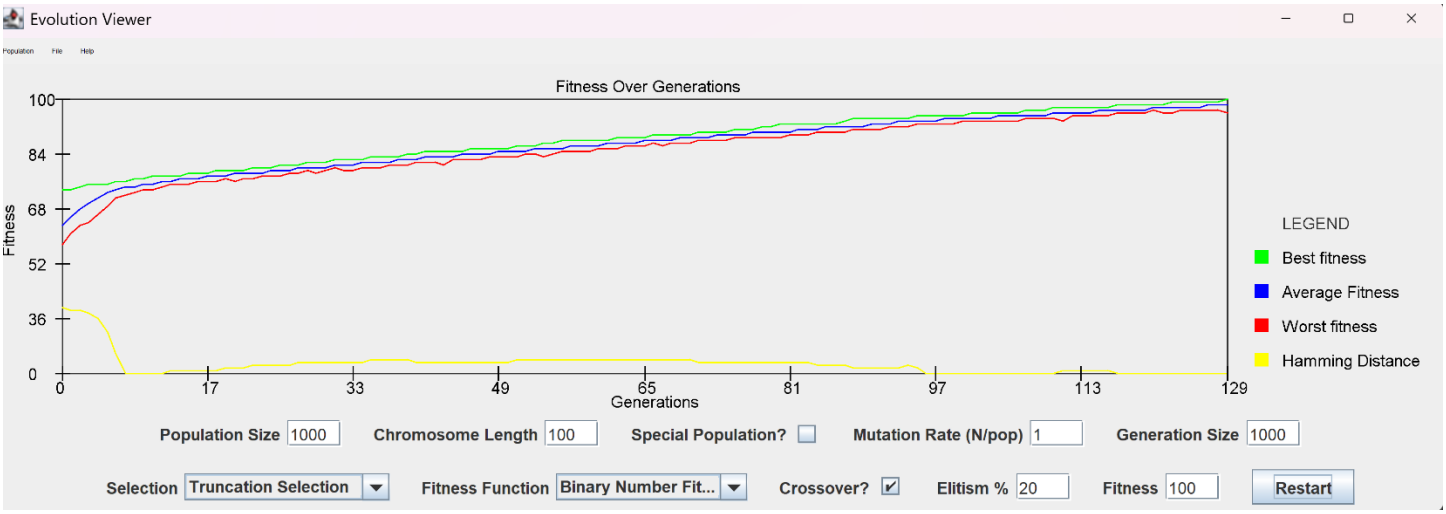
As the size of the population increased so did the number of generations needed to achieve a fitness of 10. Ther were exceptions when the population size was 10 and 50. All the graphs of the trials that reached a fitness of 10 followed the same trend. The one abnormal was when the population size was 10, which can be deduced to being caused by the small population size.




Evolution Viewer
Population File Help
Fitness Over Generations
Fitness
Generations
LEGEND
Best fitness
Average Fitness
Worst fitness
Hamming Distance
Population Size 50
Chromosome Length 100
Special Population?
Mutation Rate (N/pop) 1
Generation Size 1000
Selection Truncation Selection
Fitness Function Binary Number Fit...
Crossover?
Elitism % 20
Fitness 100
Restart


Evolution Viewer
Population File Help
Fitness Over Generations
Fitness
Generations
LEGEND
Best fitness
Average Fitness
Worst fitness
Hamming Distance
Population Size 100
Chromosome Length 100
Special Population?
Mutation Rate (N/pop) 1
Generation Size 1000
Selection Truncation Selection
Fitness Function Binary Number Fit...
Crossover?
Elitism % 20
Fitness 100
Restart


Evolution Viewer
Population File Help
Fitness Over Generations
Fitness
Generations
LEGEND
Best fitness
Average Fitness
Worst fitness
Hamming Distance
Population Size 500
Chromosome Length 100
Special Population?
Mutation Rate (N/pop) 1
Generation Size 1000
Selection Truncation Selection
Fitness Function Binary Number Fit...
Crossover?
Elitism % 20
Fitness 100
Restart



Discussion

It can be concluded that with elitism and truncation there is a minimum amount of population that is needed before it becomes efficient as seen with a population of size 10 never reaching a fitness of 10 and the population of size 50 reaching fitness 10 with a greater number of generations than a population size of 100. While not conclusive, it appears that there is a trend that all population after size 100 have an increased number of generations to reach fitness 100, but the increase in generations is not by a vast amount.

6 Discussion

[What are the takeaways from the results?]

- The takeaways from the results are the things such as crossover, i.e. the learning aspect, helps the mutation go in the wanted direction, but there still needs to be a random component such as the random mutation to achieve change rather than everything resembling the fittest initial chromosome.

[How do they connect to the ideas from your introduction?]

- These connect to our ideas from the introduction illustrating what things need to be emphasized when trying to mutate a genotype to a desired genotype.

[How does this connect to biology/engineering/art? How accurate or inaccurate is it?]

- This connects to biology through the idea of adaptation, when a species is only learning with no mutation there is a shift over generations to have the characteristics of the fittest specimen, but there is no actual genetic change in the species.

7 Conclusion

[What is the main takeaway(s) from the experiments?]

- The takeaways are the fact that when an organism's learning and mutation can take place in many ways. With different combinations of these methods of learning and mutating, the number of generations needed to achieve a particular level of fitness can be reduced significantly.

[Summarize what you learned from this project and your experiments.]

- We learned teamwork, perseverance when dealing with GitHub, problem solving, how to refactor and design code for being used in multiple instances.

8 Acknowledgments

Thank you to our TA Allyn Loyd as well as PJ Chen.

9 Works Cited

[List any papers you referenced, coding snippets, explanatory websites, videos, or other resources that were used during the project.]

Newest Questions - Stack Overflow
Oracle | Cloud Applications and Cloud Platform
GeeksforGeeks | A computer science portal for geeks
https://softwareengineering.stackexchange.com/